

Tortoises and turtles

PHYLUM	Chordata
CLASS	Reptilia
ORDER	Testudines
FAMILY	14
SPECIES	346

Tortoises and turtles are among the oldest of all living reptiles. They first appeared about 200 million years ago but have evolved little in the intervening time, so that the living species are remarkably similar to those that lived side by side with such animals as dinosaurs. Their most distinctive feature is the hard shell that encloses the

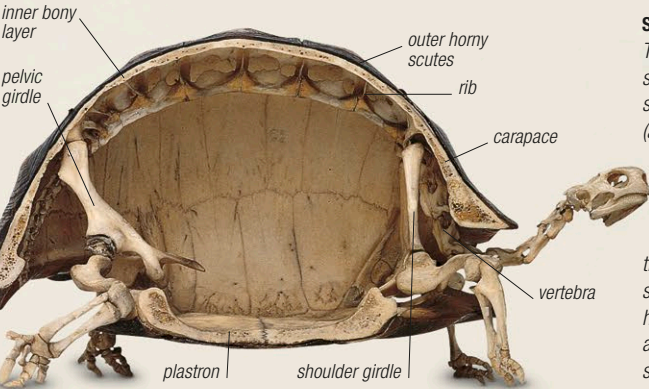
soft parts of the body, providing protection and camouflage from predators and the elements. Tortoises and turtles have no teeth and instead use sharp jaws to cut their food. They live on land as well as in freshwater and marine habitats (although all species lay their eggs on land). The terrestrial species are commonly referred to as tortoises, while those that live in fresh water are often called terrapins. The term turtle was traditionally reserved for marine species, but most zoologists now use it to refer to all members of the order. Although they are most common in tropical regions, tortoises and turtles are also found in temperate parts of the world. Some marine species undertake long-distance migrations, either in search of food or to reach their nest sites.

Anatomy

All tortoises and turtles have a shell, 4 limbs, and a horny, toothless beak in their jaw. The shell consists of upper and lower parts (known as the carapace and plastron, respectively), joined between the front and hind legs on each side by a bridge. All parts of the shell have 2 layers: an underlying bony layer and an outer epidermal layer. The outer layer is made of thin, horny plates (scutes), which contain the pigment that gives each species its distinctive coloration. Some species lack scutes and have soft, leathery shells. The shape of the limbs differs between terrestrial and aquatic species: most terrestrial species have short, club-shaped legs, while in aquatic species, they are either webbed or shaped like flippers. Since their ribs are fused to the shell (see below), tortoises and turtles cannot move their ribs to draw air into and out of their lungs. Instead, they use muscles at the tops of their legs to provide the necessary pumping action.

SKELETON

Tortoises and turtles have an unusual skeleton. The ribs and some of the vertebrae are fused to the inner surface of the carapace, and the pelvic and shoulder girdles are in an unusual position inside the rib cage. The skull is heavily built, with no openings behind the eye sockets, as there are in other reptiles. The length of the neck varies greatly between species and determines how the head is withdrawn into the shell (see right).



DOMED SHELL



STREAMLINED SHELL

SHELL SHAPES

The shape of a tortoise's or turtle's shell reflects its way of life. Terrestrial species, like the Indian star tortoise (above top), have a high, domed shell that is difficult for predators to bite or crush. Aquatic species, like the red-eared slider (above), have a low, streamlined carapace that helps them slip easily through water. A few species, such as the pancake tortoise, have a flexible carapace. Others, such as box turtles, have a hinged plastron so that they can close up completely.

FINDING THE WATER

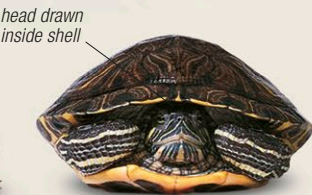
Upon hatching from their eggs, young sea turtles, such as these green sea turtles, have to find their way to water. It is thought that they either instinctively travel down the slope of the beach or sense small differences in light levels over the land compared with the water (the water being brighter by both day and night). On their way, they must contend with predators such as gulls and crabs.

SIDE- AND STRAIGHT-NECKED TORTOISES AND TURTLES

Two major subdivisions of tortoises and turtles are recognized, based on the way they withdraw their head into their shell. Side-necked turtles bend their neck sideways, so that the head rests under the edge of the shell; all side-necked turtles are aquatic or semiaquatic, freshwater species. Straight-necked species (which include all terrestrial and some aquatic forms) have a shorter neck, which they bend into a vertical S-shape inside the shell, so that their head appears to go straight back when it is withdrawn.



SIDE-NECKED



STRAIGHT-NECKED